MINIMAL NON-SCHEME QCQS SPACES ARE REDUCED AND IRREDUCIBLE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4055-4083, following the statement at 4047-4053.

Editorial note. The proof text ends at 'Thus W is an affine scheme'; the final inference (X covered by affine scheme neighbourhoods is a scheme, contradicting the hypothesis) is the standard Zariski-cover criterion and is noted.

Allowed standard prerequisites (editorial).

- Reduction of a minimal non-scheme is a scheme. lemma-reduction-scheme (same chapter, cited at ~4056). X is reduced.
- Reduced induced closed subspace. Properties of Spaces, Definition spaces-properties-definition-reduced-induced-space (cross-chapter, cited at ~4059-4061). Z_1, Z_2 corresponding to proper closed subsets are schemes.
- Pushout of schemes along closed immersions. More on Morphisms, Lemma more-morphisms-lemma-pushout-along-closed-immersions (cross-chapter, cited at ~4063-4066). Z_1 amalga_Z Z_2 exists in the category of schemes.
- Target of a finite surjection from an affine scheme is affine. lemma-affine (same chapter, card L26308, cited at ~4070 and ~4081-4082). W is an affine scheme.

Source reference. Original lemma 07VS: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 3592–3592. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07VU: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 3699–3699. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B7M: focused-seeds/geometry/S9-spaces-limits-minimal-nonscheme/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 047X: focused-seeds/geometry/S9-spaces-limits-minimal-nonscheme/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the

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counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. Assume that every proper closed subspace $Z \subset X$ is a scheme, but X is not a scheme. Then X is reduced and irreducible.*

Proof. We see that X is reduced by Lemma 07VU. Choose closed subsets $T_1 \subset |X|$ and $T_2 \subset |X|$ such that $|X| = T_1 \cup T_2$. If T_1 and T_2 are proper closed subsets, then the corresponding reduced induced closed subspaces $Z_1, Z_2 \subset X$ (Properties of Spaces, Definition 047X) are schemes and so is $Z = Z_1 \times_X Z_2 = Z_1 \cap Z_2$ as a closed subscheme of either Z_1 or Z_2. Observe that the coproduct $Z_1 \amalg_Z Z_2$ exists in the category of schemes, see More on Morphisms, Lemma 0B7M. One way to proceed, is to show that $Z_1 \amalg_Z Z_2$ is isomorphic to X, but we cannot use this here as the material on pushouts of algebraic spaces comes later in the theory. Instead we will use Lemma 07VS to find an affine neighbourhood of every point. Namely, let $x \in |X|$. If $x \notin Z_1$, then x has a neighbourhood which is a scheme, namely, $X \setminus Z_1$. Similarly if $x \notin Z_2$. If $x \in Z = Z_1 \cap Z_2$, then we choose an affine open $U \subset Z_1 \amalg_Z Z_2$ containing z. Then $U_1 = Z_1 \cap U$ and $U_2 = Z_2 \cap U$ are affine opens whose intersections with Z agree. Since $|Z_1| = T_1$ and $|Z_2| = T_2$ are closed subsets of $|X|$ which intersect in $|Z|$, we find an open $W \subset |X|$ with $W \cap T_1 = |U_1|$ and $W \cap T_2 = |U_2|$. Let W denote the corresponding open subspace of X. Then $x \in |W|$ and the morphism $U_1 \amalg U_2 \to W$ is a surjective finite morphism whose source is an affine scheme. Thus W is an affine scheme by Lemma 07VS. $\square$